ANALYSIS AND INTERPRETATION OF DEMOGRAPHY AND SOCIAL STATISTICS

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DEFINITIONS OF TERMS

1. Median age	: The age which divides the population into two equal-size groups, one of which is younger and the other older than the median.
2. Crude birth rate	: The number of live births registered in a year per 1,000 mid-year population.
3. General fertility rate	: The number of live births occurring in a year per 1,000 women aged (15-49) years at mid-year.
4. Total fertility rate:	: The average number of children born to an average woman assuming that she survives to the end of her child-bearing age and is subjected to a fixed schedule of age-specific fertility rates.
5. Gross reproduction rate	: The average number of daughters born to an average women assuming that she survives to the end of her child-bearing age and is subjected to a fixed schedule of age-specific fertility rates.
6. Net reproduction rate	: The extent to which an average woman will replace herself assuming a fixed schedule of age-specific mortality and fertility rates to prevail through her reproductive span of life.
7. Crude death rate	: The number of deaths in a year per 1,000 mid-year population.

8. Infant death	: Death of an infant aged under one year.
9. Infant mortality rate	: The number of infant deaths in a year per 1,000 live births during the year.
10. Under 5 mortality rate	: The number of deaths of children aged under 5 years in a year per 1,000 live-births during the year.
11. Still birth	: A child born after the twenty-eighth week of pregnancy and not breathing or showing any signs of life at any time after being completely expelled from the mother.
12. Maternal mortality rate	: The number of deaths due to pregnancy, childbirth and the puerperium per 1,000 live births during that year.
13. Marriage rate	: The number of persons married in a year per 1,000 mid-year population.
14. Divorce rate	: The number of persons divorced in a year per 1,000 mid-year population.
15. Enrolment rate	
(i) Primary	: The number of students (including students of less than 6 years or greater than 11 years) enrolled per 100 population aged (6-11) years
(ii) Secondary	: The number of students enrolled per 100 population aged (12-19) years.
16. Juveniles	: Persons aged 12 – 17 years.

17. Juvenile delinquency rate	:The number of juveniles in offences per 1,000 mid-year population aged 12 to 17 years.
18. Economically active population	: The economically active population is made up of the employed and the unemployed populations. The employed population consists of person, who are working while the unemployed population consists of persons who are not working but who are looking for work and are available for work.
19. Not economically active population	: The inactive population is neither employed nor unemployed and consists mainly of students, homemakers, retired persons and the disabled.
20. Government Services	: Total Government Services include both Central and Local Governments but exclude Extra Budgetary (EBUs).

Chapter 1

1.0 Introduction

At the end of the module the student should be able to:

a) Define demography with respect to its size, distribution, structure and change
b) Define nuptiality and how it relates to human population.
c) Differentiate between immigration and emigration.
d) Differentiate between de-jure and de-facto.
e) Identify the different uses of demographic data.
f) Identify different sources of demographic information.

1.1 Demography

A scientific study of human population, with respect to their: - Size, Distribution (composition), Structure and Change - caused by fertility, nuptiality, mortality and migrations.

Size: - The number of persons in the population.

Distribution: - The arrangement of the population in space at a given time (Geographically, among various residential areas or occupations).

Structure: - The distribution of the population among its sex and age grouping.

Change: - The growth or decline of the total population or one of its structural units.

1.2 Introduction

DEMOGRAPHY is a scientific study of human population primary to their size, structure and their development.

“Formal Demography” is defined by the same as: - the treatment of quantitative relations among demographic phenomena in abstraction with other phenomena. This is constructed with the broader term “population studies” which also includes

the treatment of relations between demographic events and social economic and other phenomena.

Demography studies the size of a group of people, how the number has changed in the past and the likelihood of changing in future; factors determining the changes and the composition structure of the population in terms of age, sex, age, tribe or nationality, birth place, previous residence, education, marital status etc.

Major topics covered under demography include:-

- Fertility
- Mortality
- Migration and immigration

Fertility, mortality and migration are components of population changes and affect its geographical distribution. Knowledge about them is used in population estimations and projections among other uses. One of the most important tools in Demographic studies is a life table.

NUPTIALITY varies with the age of first marriage common in a society and with the age structure of the population; obviously it will be lower in an ageing population than in a very young one. It is a major factor in fertility, although less so now with a large proportion of children born out of wedlock. The most basic rates express the number of marriages per thousand populations, or the number of people per thousand marrying in a given year.

MORTALITY - the number of people in a population dying in a year, related to two other terms: ***life span*** and ***life expectancy***. ***Life span*** is the life-length of the person who has lived longer than any other, and that is now 122 years. ***Life expectancy*** - the average longevity in a population (75 for men and 81 for women).

MIGRATION - movement of people in and out of an area or society. ***Immigration*** is movement into an area. ***Emigration*** is movement out. ***Push factors*** compel people to emigrate. ***Pull factors*** induce them to immigrate. Depending on these factors migrations can be more or less ***voluntary*** or not. Factors determining migrations are: ***economic***, primarily in terms of opportunity; ***political***, such as

persecution or subjugation/suppression; ***environmental*** or ***natural***, like disaster or famine. ***Immigration rate*** - the number of people entering an area. ***Emigration rate*** is a number of people leaving an area. ***Gross migration rate*** is immigration plus emigration rate. ***Net migration rate*** is immigration minus emigration. Migrants are people who moved - and they can be ***internal*** or ***external*** to an area.

1.2.1 The Definition of Population

The term population is used differently in demography and in statistics. In demography it relates to the number of people in a given area, while in statistical it is means the universe of units under consideration, which may be people or items.

However there are numerous ways of defining population in demographic way. This depends on whether one includes or excludes groups such as visitors, seamen, armed forces, diplomats, institutional population, those temporarily away or those missed by census.

Broadly there are two types of population counts; de facto and de Jure.

Population pyramids comprises of age sex composition;

There are about four population shapes:-

1. Pyramidal - indicating high growth
2. Rectangular - indicating slow growth
3. Oval - indicating decline
4. Hourglass - indicating increased pandemic e.g. AIDS driven decline

1.2.2 De-Facto and De-Jure

De-Jure means the populations who belong to the area or have the legal right to be there through **citizenship** and have legal evidence. In order to try to obtain some comparability between nations, the united nations recommended that each

country should produce from the census a population total which excludes foreign military and diplomatic staff currently in the country, but includes their own armed forces, merchants, seamen and diplomats who are currently abroad.

De facto means the populations who are **present** in a country at the time of census irrespective of the people's citizenship. It counts everybody so long as they are in an area at the time of counting.

1.3 Uses of Demographic Information

i. Economic Planning

Population size, growth rate and available labour force are useful factors in planning the economy of the country to sustain the population.

ii. Policy making

Taking population growth rate into account the government may make a policy to encourage family planning.

iii. Political and Administrative purposes

Whether or not to sub-divide an administrative area mainly depends on the population to be served. The number of constituencies in a given district depends on the population of a district.

iv. Distribution of social welfare

Institutions, facilities and services e.g. health, education, security, water supplies, transport, etc depend on the population of a given area.

- The numbers of Health Institutions to be established in a given district depend on the population to be served.

- The number of primary schools to be established in a district depends on the number of children of primary school going age.
- The capacity of water supply installation to provide an urban centre depends mainly on the population of the centre apart from other factors.

v. Health Indicators

The health status of a community is easily portrayed by various rates calculated in demography e.g. mortality rate, fertility rates, morbidity rates, birth rates and life expectancy.

vi. Social Research

Any social research must make use of some demographic information. Demographic information is also useful in the formulation of health policies and medical research.

vii. Distribution of communal burdens

A country like Kenya which has relied on fund raising (Harambee) for development of most of the social infrastructures like schools, and health institution has to allocate quotas to divisions, locations, sub-locations or villages mainly according to their population.

viii. Distribution of Human Resources

To determine human resources in terms of labour as a factor of economic production, its potential and actual exploitation (labour force, employed, underemployed, unemployed).

1.4 Sources of Demographic Data

The raw materials of formal demography are data collected from:-

i. Population census
ii. Demographic sample surveys
iii. Registration of vital events (births, deaths, migration and marriages)
iv. Registration of persons (e.g. I.D)
v. Hospital records
vi. Various publications e.g. UN statistical year book
vii. National Social Security Scheme.

Questions for Chapter One

1. What is the scientific definition of “demography”
2. What is nuptiality and how does it relates to human population.
3. Differentiate between “migration in” and “migration out”.
4. With example, differentiate between “de-jure” and “de-facto”.
5. State at least five (5) uses of demographic data.

Chapter 2

2.0 Introduction

At the end of the module the student should be able to:

a) Define demographic measure used in demography
b) Apply the basic demographic equation

2.1 Demographic Measure

Ratios, Rates and Proportions are the basic tools of formal demography. Essentially they enable comparison by eliminatory differences due to population size.

A ratio is simply any number divided by any other number. A proportion is a type of ratio that is by the formula:-

$$\text{Proportion} = \frac{x}{x + y}$$

A proportion can only range from 0.0 to 1.0. It is though very common to express proportions as percentages simply by multiplying by 100 in which case the range of possible value is from 0% to 100%. The term “rate” is used very loosely in demography and can course confusion. Strictly, the numerator of the rate is a number of events, such as births or deaths occurring during a period of time, and the denominator is the population at risk of experiencing these events. The crucial point is that some time period has to be specified. More precisely the denominator is the № of “persons- years of exposure to risk” experienced by the population during the period under consideration. In demography, rates are most frequently calculated by periods of one year.

This means that the total number of persons- year of exposure to risky can usually be approximated by the population at the mid- year. To understand this, consider the calculations of an annual death rate. Each person servicing for the whole year will contribute one year to the total number of years of exposure to risk, while those dying during the year will contribute only a fraction of the year. This fraction will be on average half a year if people die evenly throughout the year. The total number of years of exposure to risk arrived at by adding these fractions to the total contributed by those who survived is clearly the same as taking an average population total during the year and letting each and every person in that average population – contribute a whole year of exposure. And since the average population can normally be approximated very closely by the population at the mid-year, the mid-year population can under most circumstances, be used for calculations of annual rates. This is why most governments statistical services produce population estimates relating to the middle of each year, rather than the beginning or end.

2.2 The Basic Demographic Equations

This is one of the most fundamental relations in demography. It can be written as:-

$$Pt_1 = Pt_0 + (B - D) + (IN - OUT\ MIGRATION)$$

Pt_1 = Population at the end of the period.

Pt_0 = Population at the beginning of the period.

B = Births during the period.

D = Deaths during the period.

IN = Migration IN (immigration) during the period.

OUT = Migration OUT (emigration) during the period.

The difference between the number of births and deaths gives the "**natural increase**" whiles the difference between the numbers of migrants in and out gives the "**net migration**". Thus the equation can be written as:-

Population change = natural increase + net migration.

The equation can also be expressed in terms of rates by dividing each element by the mid-year population (MYP). Then the rate of population change is just

$$\frac{Pt_1}{(MYP)} = \frac{(B - D)}{(MYP)} + \frac{(IN - OUT)}{(MYP)}$$

the difference between the birth rate & death rate is just the rate of natural increase and the difference between the in and out migration rates are the net migration rate so the equation can be written as:-

Natural increase = (B –D)/MYP x 100

Net Migration = (IN – OUT)/100 x 100

Growth rate = natural increase rate + net migration rate

2.3 Migration IN Rate

The migration IN rate (immigration) is the rate at which people get into a country or region per a hundred people i.e. MI/MYP x 100. Migration OUT (emigration) rate is the rate at which people are getting out of a country i.e. MO/MYP x 100.

2.3 Doubling point of a population

Doubling point of the population is the period that the population is expected to double from its current times 2. For example if the population is 28 million, when is it expected to double to 56 million? The formula for doubling point of the population is 70/R. 70 is constant and R is the growth rate (%)

Example 1;

Kenya growth rate in 1997 was 3.3%, present population in 1997 was 28 million. When will it double to 56 million?
70/3.3 = 21.27 years. Add 1997 + 21.27 = 2018. Therefore, the population is expected to double in the year 2018 from 28m to 56m.

Example 2:
Can also use the following formula;-

$Pt_1 = Pt_0 (1 + r)^t$

Pt_1 = Population at the end of the period

Pt_0 = Population at the beginning of the period

r = growth rate

t = time (period)

Thus using the example above, calculate when the population will double:-

$Pt_1 = Pt_0 (1 + r)^t$

$= 28(1 + 0.033)^{21}$

$= 55,370,000$

2.4 Population decay

Each year the local country club sponsors a tennis tournament. Play starts with 128

participants. During each round, half of the players are eliminated. How many players remain after 5 rounds?

Rounds	1	2	3	4	5
Number of Players left	64	32	16	8	4

There are 4 players remaining after 5 rounds.

Function: $y = a(1-r)^x$	a = the initial amount before the decay begins r = decay rate x = the number of intervals
$y = 128(1-.50)^x$	as x ranges from 1 to 5 for this problem

$Pt_1 = Pt_0 (1 - r)^t$

$= \quad 128(1 - 0.50)^5$

$= \quad 4$

Only 4 players will remain after the 5th round.

Question for the module

1. In Bukura constituency, there were 151, 326 people at the beginning of the period in April 1950. Births that occurred during the period from April 1950 to 31st March 1960 was 40, 963. Deaths that occurred during the said period were 15, 608 and the Net Migration (IN – OUT Migration) was 2,645. Using the information above, calculate the population at the end of the period.
2. A bank account balance, b, for an account starting with s dollars, earning an annual interest rate, r, and left untouched for n years can be calculated as $b = s(1+r)^n$ (an exponential growth formula). Find a bank account balance to the nearest dollar, if the account starts with $100, has an annual rate of 4%, and the money left in the account for 12 years.

Chapter 3

3.0 Introduction

At the end of the module the student should be able to:

a) Define population census
b) Describe civil registration

3.1 Censuses

A U.N publication lists four essential features of a population census.

- Each individual is enumerated separately and the characteristics of each person are recorded separately.
- The census covers a precisely defined territory and includes every person present or residing within its scope.
- The population is enumerated with respect to a well defined point of time and data are in terms of a well defined reference period.
- Censuses are taken at regular intervals.

3.2 Sample Surveys and Registration Systems

Current sample surveys like censuses represent periodic stock-takings whereas registration systems and immigration and emigration control systems, involve the continuous recording of data even though the published statistics are extracted only at specified intervals.

3.3 Sample Survey

Many surveys are undertaken by private organizations, periodic sample surveys constitute a continuing micro-census of a country or other area. The scope of the survey and the methods of recording the data should be subject to the same principles as a census.

3.4 Registration Systems (Requirements)

The equivalent of an “individual enumerator” is the individual vital record.

Procedures to assure the filling of a uniform record for every vital event for example live birth, death or marriage.
Procedures to provide for complete and unable answers to the inquiries on the record form.
Procedures to enable information in the record form to be processed for statistical purposes through some central office which provides vital statistics for the nation and its sub-divisions on a comparable.
Specified maximum interval of time between the occurrence of an event and its recording.
The date-day-month-year-of both occurrence and registration should be on the form.
Compilers of statistics from registers like the agencies collected and tabulated census data, are obligated to publish, evaluate and to some extend analyze.

3.4.1 Civil Registration/vital registration

This is the registration of vital events. Vital events includes the registration of births, deaths, marriages, divorce, separation, adoption, legitimation, recognition etc. registration of such data is provide key information as to changes that would take place in the population in regard to people getting into an area and departing from life. The most important things to consider in civil/vital registration is the registration of births and deaths.
Vital data or data collected from vital events can be used in many aspects in demography, demographic related research activities, administration and planning for social and economic development.
The major source of vital statistics of demographic data is vital registration or civil registration.
Vital registration systems consist of the registration of births, deaths, marriages and other vital events.
The major purpose of registration is to provide legal proof of each persons identity, indicate the date of birth of individuals (age), confirm citizenship of persons, confirm civil status of individual among others.

It is said that registration systems should be compulsory, comprehensive and permanent systems by governments for the system to provide necessary data or statistics that can be trusted. The system must be operated properly and effectively i.e. all data must be captured from the time a child is born to death.
It was noted that factors that affect civil registration are agravian societies, widespread illiteracy and shortage of adequate administrative personnel who are charged with the responsibility of capturing data at the time of need.
Late registration of births and deaths should be made easy by governments so that enough data is collected. The procedure for late registration of birth and death must not be complicated.

3.5 BIRTH REGISTRATION

Overview

The right to a name and nationality is well established. However, in 2000 alone, some 50 million births went unregistered – over 40 per cent of all estimated births worldwide that year. These unregistered children are almost always from poor, marginalized or displaced families or from countries where systems of registration are not in place or functional.

Globally, **South Asia** has the largest number of unregistered children, with approximately 22.5 million, or over 40 per cent of the world's unregistered births in 2000. In **sub-Saharan Africa**, 70 per cent of all births went unregistered in 2000. In **South Asia,** the figure was 63 per cent. In the **Middle East** and **North Africa**, nearly one third of the children born in 2000 were unregistered, while in **East Asia** and the **Pacific**, 22 per cent of births were not registered.

Apart from being the first legal acknowledgement of a child's existence, the registration of births is fundamental to the realization of a number of rights and practical needs including:

- Providing access to health care;

- Providing access to immunization;
- Ensuring that children enroll in school at the right age;
- Enforcing laws relating to minimum age for employment, handicapping efforts to prevent child labour;
- Effectively countering the problem of girls forced into marriage before they are legally eligible, without proof of age;
- Ensuring that children in conflict with the law are given special protection, and not treated (legally and practically) as adults;
- Protecting young people from under-age military service or conscription;
- Protecting children from harassment by police or other law enforcement officials;
- Securing the child's right to a nationality, at the time of birth or at a later stage;
- Protecting children who are trafficked, and who are eventually repatriated and reunited with family members;
- Getting a passport, opening a bank account, obtaining credit, voting or finding employment.

In addition to issues relating to protection, a functioning system of birth and civil registration ensures that the country has an up-to-date and reliable database for planning. This is as useful for national level planning as it is for local government bodies that are responsible for maintaining education, health and other social services for the community.

Most countries have a legal provision for registering births of children within a prescribed period. However, these laws are often not comprehensive enough, are not enforced or do not function. In many developing countries, birth registration systems have fallen into disuse. In some cases this may be due to bureaucratic lethargy and a lack of oversight. It may also be linked to a lack of resources, given the costs attached to providing birth registration. There are also practical problems, for example where births occur away from registration

locations. This includes births in isolated rural areas, or births away from medical facilities where registration normally takes place.

Sometimes there may be a deliberate element to a lack of birth registration, with particular groups being excluded due to discriminatory policies intended to minimize the size of ethnic minorities for political reasons or avoid the provision of assistance to immigrants.

Particularly in remote areas, parents often do not see the benefits of their own citizenship, let alone the benefits that birth registration would confer on their children. Where registration facilities are difficult to access or have costs attached, parents may be reluctant to register their children. A lack of parental enthusiasm for birth registration can undermine efforts to improve birth registration systems.

Building a protective environment for children

The first requirement is that birth registration be universal and free. In some countries a fee is levied. Governments have to be committed to make the resources available to register every child without discrimination. This includes organizing awareness-raising and promotion campaigns. Birth registration also requires trained staff with the necessary equipment to provide registration where people need it.

Families and communities need to be made aware of the importance of birth registration, and registration should be compulsory. Sometimes parents and communities either fail to appreciate how important it is for their child to be registered or are suspicious about the motives behind it. The media can play an important role in encouraging parents to register their children.

Explaining the value of birth registration to older children who have not been registered can enable them to encourage their parents to ensure their late

registration. It can also help ensure that they will register their own children when they themselves become parents.

Local registration facilities need to have adequate capacity to meet needs. They need to be decentralized and accessible to communities. Where registration facilities are remote there are often low levels of birth registration.

Birth registration is in itself a foundation for monitoring and reporting. It is important that birth registration data be disaggregated by gender. It is also important that birth registration cover hard-to-reach populations.

UNICEF's response

UNICEF sees birth registration as a priority in the package of services that are provided for children at birth. UNICEF seeks free birth registration, and a free birth certificate, for every child in every country. It calls for effective registration systems that are compulsory, universal, permanent and continuous, and that guarantee the confidentiality of personal data. UNICEF's approach to improving levels of birth registration focuses on:

- **Advocacy**. UNICEF supports campaigns to encourage parents to register their children. UNICEF also works closely with governments to encourage greater attention to, and resources for, birth registration;
- **Situation analysis.** UNICEF seeks to develop a clear picture of which children are not registered, where they are and why they are not registered;
- **Capacity-building.** UNICEF provides support to systems for birth registration. This ranges from training to the provision of essential materials;
- **Working with partners.** UNICEF works with a range of partners, particularly NGOs, in the promotion of birth registration;

- **Integrating birth registration into immunization programmes.** UNICEF looks for opportunities to advance birth registration through other areas of its work. Immunization is often a convenient moment for birth registration.

UNICEF in action

In **Bangladesh,** UNICEF has supported the government with a district-based approach to birth registration promotion that has now spread to include 12 districts and four large city corporations. Nearly 5 million children under six have had their births registered. This work is combined with communication activities directed at local and religious leaders, teachers and others. In rural areas, field health workers collect birth data for the registrars. In urban areas, computerized birth registration is being introduced, linking the electronic birth register with an electronic child immunization register.

In **Colombia**, UNICEF has supported the registration of displaced children in frontier zones and minority communities. At the same time, mobile registrars have been set up in rural areas where low levels of birth registration have been identified.

In the **Dominican Republic**, UNICEF has worked with the Catholic Church in 11 dioceses to promote the registration of children, including older children who were bypassed as infants.

In the **Philippines**, UNICEF has supported a range of birth registration activities including:

- Declaring a ‘Civil Registration Month’;
- Convening a national birth registration meeting;
- Strengthening civil registration systems;
- Holding biannual meetings of local registrars;
- Amending the registration law;
- Developing a central database on civil registration.

In Viet Nam, UNICEF has worked with the government and non-governmental partners in a programme including:

- Assisting in the development of a national action plan aimed at ensuring birth registration of all children in the country;
- Developing the 1998 Government Decree on Civil Registration, which establishes children's right to registration;
- Assisting in training civil registrars at district and commune/ward levels on children's rights and government regulations on birth registration.

An increase in birth registration from 60 per cent to 87 per cent has been reported from 1998 to 2001.

At a regional level, in 2002, UNICEF brought together ministers, other government officials and UNICEF staff from around the world in a consultation in Uganda to discuss how to make birth registration a higher priority for governments. Representatives from **Eritrea, Ethiopia, the Gambia, Ghana, Kenya, Namibia, Sierra Leone, South Africa, the United Republic of Tanzania** and **Uganda** shared challenges and strategies.

Birth registration rates lowest in rural areas

Percent point difference between rural and urban birth registration rates

Countries	**% Urban**	**% Rural**	**Difference**
Niger	85	40	
Chad	53	18	
Myanmar	87	53	
Senegal	80	46	
Côte d'Ivoire	88	60	
Indonesia	47	20	

Lao P.D.R.	82	56	
Sierra Leone	66	40	
Central African Rep.	88	63	
Kenya	81	57	
Vietnam	91	68	
Cameroon	94	73	
Tanzania	22	3	
Dominican Republic	82	66	
Botswana	65	52	
Gambia	37	29	
Philippines	88	81	
Georgia	97	92	
Burundi	48	43	
Comoros	87	83	
Bolivia	83	80	
Nepal	37	34	
Tajikistan	77	74	
Azerbaijan	98	96	
Bosnia & Herzegovina	98	99	
Lesotho	41	53	
Guinea Bissau	32	47	
Swaziland	50	72	

Question at the end of the module

1. Describe the process of conducting census in the population.
2. Describe the importance of civil registration

Chapter 4

4.0 Introduction

At the end of the module the student should be able to:

a) Describe the process of demographic transition theory

4.1 Demographic Transition Theory

Demographic transition emphasizes the strong interrelationships between population structure, fertility, mortality and migration which lie at heart of formal demography.

The so called theory of demographic transition was first described in the 1940's added to and rewritten many times until now there is no single, clear, generally accepted view of precisely what it comprises.

The classified description of the theory as set by NOSTEIN (1945) BLACKER (1947) and others is roughly as follows. There are a series of stages during which a population moves from a situation where both mortality and fertility are high to a position where both mortality and fertility are low.

Both before and after the transition population from growth is very slow, in between, during the transition population growth are very rapid, essentially because the decline in mortality occurs before the decline in fertility.

The theory can be disputed on several grounds both of a theoretical and on a descriptive nature. It has been criticized because it is unclear whether it is purely descriptive or whether it also includes statements about the actual cause of the transition.

The early writers about the theory differ in the emphasis they place on causal factors. Because it is formulated at a very general level it is very difficult to generate from its hypothesis which can be tested or refuted convincingly, a requirement for any scientific theory. It has also been criticized by CODE (1973) and others, on descriptive grounds. In many parts of Europe the decline in fertility occurred either before or simultaneously with the decline in mortality. The best

known example of this is France where fertility fell to a low level early on before any substantial decline in mortality.

Despite these criticisms, demographic transition theory remains at the core of demography as an academic discipline partly because of the lack of anything better to replace it.

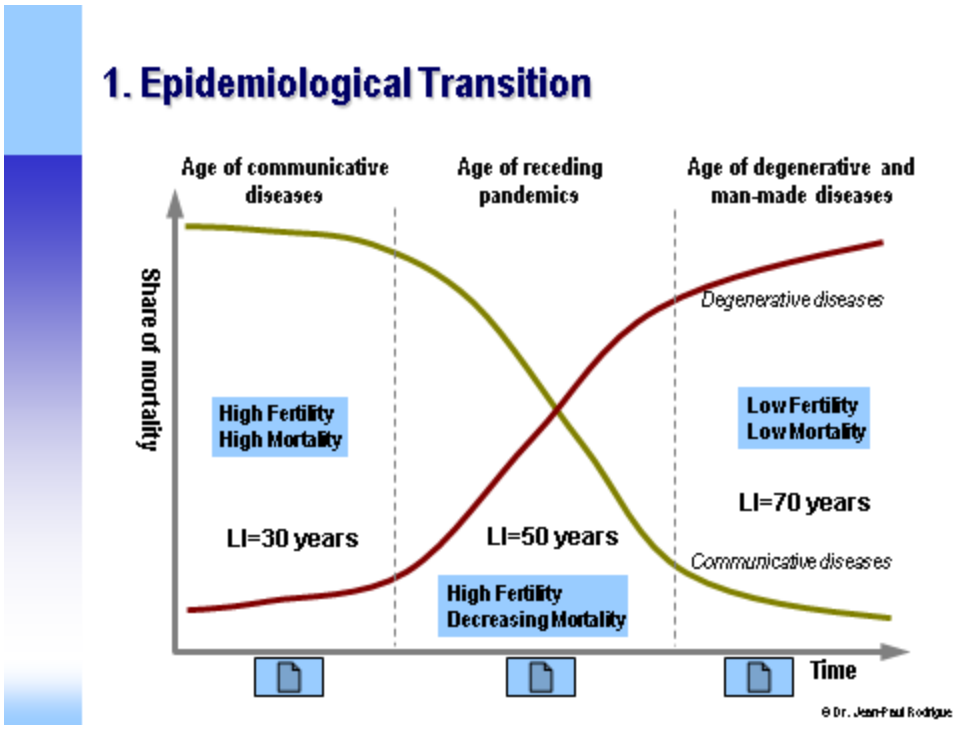
As a result, it has had a very great influence on the way the subject has developed. One particular detrimental influence is that it has demographers to consider the subject or migration as not being a part of demography.

Yet for the most practical purposes, esp. population estimates and projections, and for explaining both current and historical trends in population growth and development, migration is an essential ingredient.

One only has to consider the growth of the United States of America or Australia during the 19th century, the current population of London the depopulated rural areas of Ireland or the rapid growth of urban areas in developing countries to realize the crucial impact of migration.

Phases of Demographic Transition

Birth Rates	Death Rates	Natural Increase	Example
High	High	Zero to very slow	Europe prior to 15th century, most other developing countries till the 19th century
High	Declining	Slow	Europe 17th century, India 1930s – 40s
High	Low	Rapid	Europe 18th century, Pakistan 1970s
Declining	Low	Slow	Europe 19th century, East Asian countries in the mid 20th century
Low	Low	Zero or very slow	Europe, Japan & USA



Question of the module

a) Discuss the process of demographic transition theory

Chapter 5

5.0 Introduction

At the end of the module the student should be able to:

a) Define sex ratio
b) Define dependency ratio
c) Calculate the dependency ratio
d) Interpret the dependency ratio

5.1 <u>The Sex Ratio</u>

The overall sex ratio is simply the ratio of males to females in the population. It is normally expressed as the number of males per 100 females i.e. sex ratio = males x 100 females. The overall sex ratio is entirely determined by: -

- The sex ratio at birth

- Sex differences in mortality
- Differential migration

Sex ratio is normally around 105 boys are born for every 100 girls but it does vary between population and subgroups.

Note that if the sex ratio at birth is 105, then after every 205 births on average 100 are females i.e. the proportion that are female is thus

$\frac{100}{205} = 0.488$ males = 0.512

Sex difference in mortality and migration cause the sex ratio to very between ages. In most parts of the world, female mortality is lower than male mortality at all ages women live longer on average than men. Thus the sex ratio tends to decline with age

5.2 Dependency Ratio

One frequency used index summarizing on age distribution is known as the dependency ratio; strictly, this is the ratio of economically active to economically inactive persons in a population. However, because of lack of data or difficulties in defining economic activity in many countries a ratio of age group is used instead. Then the ratio is just:

$$\text{Dependency ratio} = \frac{(\text{Children + Elderly})}{\text{Working Ages}} \times 100$$

The praise age used depends very much on the population being studied, probably school leaving age 18 years in Kenya and retirement age of 55 years.

Question for the module

1. Define what is meant by dependency ratio
2. If there were 84 males and 80 females in a class at a college;
   a) Calculate the sex ratio of the class
   b) Interpret the sex ratio of the class.

Chapter 6

6.0 Introduction

At the end of the module the student should be able to:

a) Define fertility
b) Define fecundity
c) Calculate Age Specific Fertility Rate
d) Calculate Total Fertility Rate
e) Calculate other measures of fertility.

6.1 Period Fertility

This section defines the concept of fertility and fecundity. It then defined the commonly used measures of period fertility – the child/woman ratio, crude birth rate, general fertility rate and total fertility rate.

6.2 Fertility and Fecundity

Fertility relates to the number of live births a woman has actually had. Fertile means that a woman is or was a mother.

The physiological ability to bear children on the other hand is known as fecundity. Fecund is thus the opposite of sterile. It is thought that the maximum number of children an average woman can theoretically produce is about 15 if she starts child bearing as soon as possible after menarche, which occurs around ages 12-14 and continues until menopause in her middle or late forties.

6.3 Fertility

The following table relates to number of women and births that had occurred at least 12 months before a survey was carried out in 1979 in Kenya. Mid year population was 76,398,000.

Table 1 Women and birth 1979

Age group	Women	Births	ASFR/1000
15-19	3,777,000	756,900	200.4
20-24	3,101,000	1,046,000	337.3
25-29	2,636,000	819,000	310.7
30-34	2,161,000	565,000	261.5
35-39	1,793,000	353,000	196.9
40-44	1,484,000	141,000	95.0
45-49	1,222,000	16,500	13.5
Total	16,174,000	3,697,400	1415.3

Using the above information, calculate to one decimal place

a) A.S.F.R
b) Total Fertility Rate
c) Crude Birth Rate

a) ASFR= $\frac{\text{Birth in a year to women aged x}}{\text{Women at mid year aged x}}$ * 1000

b) T.F.R = $\frac{\sum ASFR}{1000}$ x 5

= $\frac{1415.3}{1000}$ x 5

= 7.1

= 7

CBR = $\frac{\text{Births}}{\text{Population}}$ x 1000

= $\frac{3697400}{76398000}$ x 1000

= 48.4

The following data refers to an age specific births of women in a Kenyan community.

Table 2 Age Specific births of women in Kenya

Age in years	No. of women	Births	ASFR per 1000
15-19	106400	34341	322.8
20-24	895300	162879	182.0
25-29	1358100	200167	147.4
30-34	1541500	107188	69.5
35-39	1243300	25604	20.6
40-44	1189500	5123	4.3
45-49	1196600	479	0.4
TOTAL	7530700	535781	747

Calculate ASFR

TFR for this community

$$TFR = \frac{747}{1000} \times 5$$

$$= 3.7$$

6.4 Period and Cohort Analysis

Some women are for various reasons unable to bear any children, while the greatest number ever officially recorded is 69 by a woman living near Moscow during the 18th Century (16 pairs of twins, 7 sets of triplets and 4 sets of quadruplets). The highest recorded average for any populations is about 10 births per woman among the HUTTERITES.

There are broadly two ways of approaching the study of fertility period and cohort. The essential quality of period fertility analysis is that it looks at fertility

cross sectional i.e. at births occurring during a specified period of time- normally one year cohort analysis on the other hand looks at fertility longitudinally i.e. at all births occurring to a specific group of women normally all those born or married during a reproductive history. In this study, we shall consider period analysis only because it is more frequently used.

The measures of period fertility are as follows: -

(i) Child Woman Ratio

This can be defined as

$$\text{C/W Ratio} = \frac{\text{Children Aged 0-4}}{\text{Woman aged 15-44/9}}$$

The usefulness of ratio arises from the fact that it requires only information on the age and sex structure of the population. No data on birth are needed at all. This makes it particularly useful when using census data.
Obviously it is an extremely crude measure but broadly if fertility is high, the child/woman ratio will be high while if fertility is low, the ratio will be low. It is however, quite sensitive to the level of infant mortality. So if it is dangerous to use the ratio to compare populations, with substantially different levels of infant and child mortality or where under reporting of young children is a problem.

(ii) Crude Birth Rate

This is the simplest and most frequently used as a measure of fertility. It is defined as: -

$$\text{CBR} = \frac{\text{Birth in year}}{\text{Population at Midyear}} \times 1000$$

It is always expressed per 1000 population. The reasons it is a “crude” rate is that it includes all ages and both sexes in the denominator.

No attempt is made to relate the births to the women at risk of having those births because of this it is strictly not a measure of fertility at all.

The range of CBR is from 10 - 50. A major problem is that it is affected very greatly by the composition of the population as regards age, sex and other characteristics. Thus it can be very misleading if it is used to compare different populations or the same population at different times because it may vary greatly in their composition. There are though 3 reasons why it is a useful measure: -

It is easy to understand.

It requires few data and it is easy to calculate - all that is required is the total number of births and the total population.

It is possible to subtract the crude death rate from the crude birth rate to get the crude rate of natural increase, which along with the crude net migration rate can estimate, the present population. This refers to demographic equation.

Table 2 Age Specific Fertility Rate Bangladesh 1974

AGE GROUP	WOMEN	BIRTH	ASFR PER 1000
15-19	3777000	756900	200.4
20-24	3101000	1046000	337.3
25-29	2636000	819500	310.9
30-34	2161000	565100	261.5
35-39	179300	35320	197.0
40-44	148400	14160	95.4
45-49	1222000	16500	13.5
TOTAL	**13224700**	**3253480**	**1416.0**

Population = 76398000

Women (aged 15-49) = 13224700

Births = 3253480

$$\text{Crude Birth Rate} = \frac{3253480}{76398000} \times 1000 = 42.59/1000$$

6.5 General Fertility Rate

This is a fertility rate rather than a birth rate because it expresses the births relative to the No. of women of the reproductive age. Sometimes the denominator includes women aged 45-49. Sometimes it doesn't. This is important because compiling a 15-44 GFR with 15-49 GFR could be misleading.

E.g. the Bangladesh 1974 ages 15 -44.
GFR of age 15 -44/9

$$\text{GFR} = \frac{\text{No. of Births during the year}}{\text{Women aged 15 – 44/9 at midyear}} \times 1000$$

$$\text{GFR of age 15 – 49} = \frac{3253480}{13224700} \times 1000 = 246.02/1000$$

Using the same data you now calculate the GFR 15 – 44 and GFR 15-49. The possible value is roughly 50 -300. Note that more data are required to calculate the GFR than CBR. One has to know the age and sex composition of the population. The main advantage of GFR over CBR is that it substantially controls for age and sex structure by relating the births to roughly the women at risk of having them, but it does not control entirely for age structure since there may be substantial variations between populations within the reproductive age range. For example in one there may be a relatively large number of girls aged 15 – 19 while in another there may be only a few. Because births are not spread evenly over the reproductive age range, comparisons using the GFR can be misleading. To overcome this it is necessary to move to a substantially more sophisticated level and consider fertility for each age group.

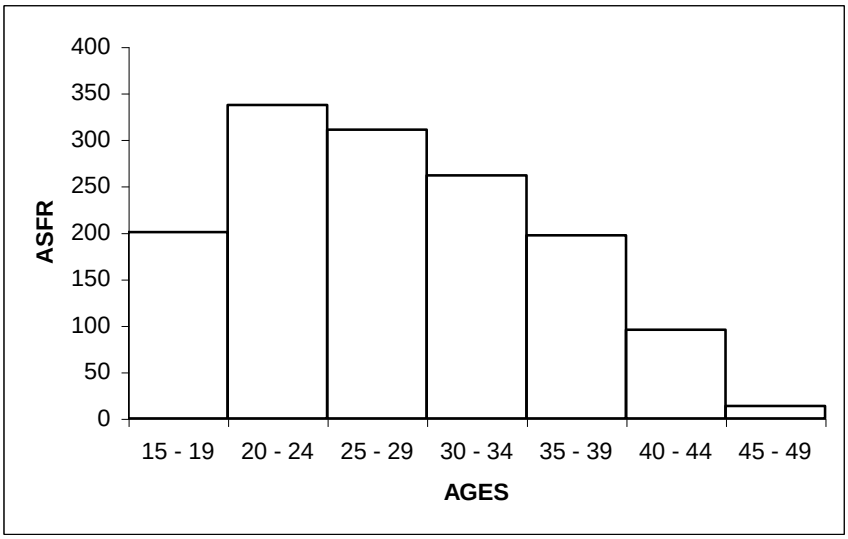
6.6 Age Specific Fertility Rate

$$\text{ASFR} = \frac{\text{Births in a year to women aged } x}{\text{No of women aged } x \text{ at midyear}} * 1000$$

$$\text{ASFR (15-19)} = \frac{756900}{3777000} \times 1000 = \mathbf{200.39}$$

Often ASFR are expressed per 1000 women normally 7 age groups, which has a class interval of five years. They normally start with ages 15 – 19 and go on to 45 – 49. Table No.3 shows the age groupings the women, births and ASFR for Bangladesh 1974. Note that there is a big jump over the measures considered so far. It is necessary to have births classified by age of mother.

Figure 1 Age Specific Fertility Rate Bangladesh



ASFR can be graphed in figure 1. It shows features such as rapid rise into a peak in the early or mid 20s and a gradual decline to very low levels after age 40. It is conventional to include any births occurring to girls aged below 15 in the 15-19 age groups on the grounds that their ages are being underestimated. Similarly births of women over 49 years are usually added into the 45- 49 group. The effects of these are usually negligible since there are very few births in these ages. The great inconvenience of Age Specific Fertility Rate (ASFR) is that they are not a single number but a set of 7 (seven). This makes comparisons

complex and tedious but fortunately it is possible to overcome this by using the Total Fertility Ratio (TFR).

6.7 Total Fertility Rate

It is probably the measure of fertility that is most widely used by demographers. It is calculated by adding up the ASFRs. Each ASFR usually relates to 5 years. It is therefore necessary to multiply the 5-year rate by five. The TFR is almost always expressed per one (1) woman where as ASFRs are often expressed per 1000 women. If that is the case then it is necessary to divide by 1000.

$$TFR = \frac{\text{Sum of ASFR}}{1000} \times 5 \quad \text{or} \quad \frac{\Sigma ASFR}{1000} \times 5$$

Example Bangladesh 1974

$$TFR = \frac{(200.4 + 337.3 + 310.9 + 261.5 + 197.0 + 95.4 + 13.5)}{1000} \times 5$$

$$= \mathbf{7.08}$$

Three points to note when computing TFR are: -

If single year ASFR are used then there is no need to multiply by 5. The 5 are used because the age interval is 5.

It makes no difference if you multiply the ASFRs by 5 and then add them or the other way round except that it is easier to add them first.

It is perfectly acceptable to include the ASFR for the age 10-14 or 50 – 54 or omit 45 – 49. However, they are so small that they make very little difference to the final figure.

The great value of the TFR is that it is single figure measure that is independent of age structure. The GFR only partially controls for age structure and the CBR doesn't do that at all. One problem with TFR however, is that it requires a lot of

data – births by age of month and women by age group which is usually only available where there are high quality registration and census systems.
Another problem is its interpretation. Formally, the TFR can be thought of as the number of children a woman would have if: -
She survived to age 50.
Throughout her reproductive life she experienced exact ASFR for the year in question.

To understand this considers a group of women in Bangladesh aged 15. While they are 15 – 19 they have an ASFR of 200.4 then 337.4 while they are 20-24 and 13.5 from 45-49. If they all survived to age 50 they will have had an average 7.08 children each.

6.8 Age Specific Fertility Rate

Ordinary ASFR are calculated by dividing the births occurring to women aged x by the total No. of women aged x.

$$\text{ASFR} = \frac{\text{Births occurring to women aged } x}{\text{Total No. of women aged } x} \times 1000$$

However it is also possible to define fertility rates that are specific to many other things, almost always in conjunction with total status. The following are the measures of marital status: -

General marital fertility rate

$$\text{GMFR} = \frac{\text{All Births}}{\text{Married women 15-44/49}} \times 1000$$

General legitimate Fertility Rate

$$\text{GLFR} = \frac{\text{Legitimate Births}}{\text{Married women 15-44/49}} \times 1000$$

General Illegitimate Births

$$GIB = \frac{\text{General Illegitimate Births}}{\text{Single, widowed, divorced, and separated Women aged 15-44 or 49}} \times 1000$$

These are the measures of marital fertility.

Marital Age Specific Fertility Rate:

Sometimes is also called Age Specific Marital Fertility Rate. This is simply obtained by calculated Age Specific Fertility Rate. Just the way fertility rate was calculated.

Table 3 Marital Age Specific Fertility Rate for England 1978

Age Groups	Legitimate Birth	Married Women	MASFR
15-19	34341	106400	322.8
20-24	162879	895300	181.9
25-29	200167	1358100	147.4
30-34	107188	1541500	69.5
35-39	25604	1243300	20.6
40-44	5123	1189500	4.3
45-49	479	1196600	0.7
TOTAL	**535781**	**7530700**	**747.2**

The MFR just as with the TFR is necessary to multiply by 5 if using 5-year age groups divide by 1000 if the MASFR's being used are expressed per 1000 women.

$$MFR = \frac{\Sigma MASFR}{1000} \times 5$$

$$= \frac{747.2}{1000} \times 5 = \mathbf{3.736}$$

Table 4 Legitimate and Married women births

Age Groups	Legitimate Birth	Married Women	MASFR
15-19	24221	53400	453.6
20-24	152778	795200	192.1
25-29	100150	1257100	79.7
30-34	55182	1420300	38.9
35-39	22504	1110200	20.3
40-44	5230	101130	51.7
45-49	500	1030500	0.5
TOTAL	**360565**	**5767830**	**836.8**

To calculate MASFR

$$\text{MASFR} = \frac{\text{No. of births of women with x ages}}{\text{No. of women of the same age group}} \times 1000$$

15 –19 = $\frac{24221 \times 1000}{53400}$ = 453.6

20 - 24 = $\frac{152778 \times 1000}{795200}$ = 192.1

25 - 29 = $\frac{100150 \times 1000}{1257100}$ = 79.7

30 - 34 = $\frac{55182 \times 1000}{1420300}$ = 38.9

35 – 39 = $\frac{22504 \times 1000}{1110200}$ = 20.3

40 - 44 = $\frac{5230 \times 1000}{101130}$ = 51.7

45 – 49 = $\frac{500 \times 1000}{1030500}$ = 0.49 or 0.5

$$MFR = \frac{\Sigma MASFR}{1000} \times 5$$

$$= \frac{836.8}{1000} \times 5 = \mathbf{4.184}$$

Question for the module

From the following table relating to the number of women and births which occurred in a District in Tanzania in 1989. Population of the District was reported at 1,467,450

Age groups	**No. of women**	**No. of births**
15 - 19	95,890	15,350
20 – 24	145,640	39,680
25 – 29	125,580	28,960
30 – 34	87,900	15,860
35 – 39	79,840	10,680
40 – 44	51,380	4,560
45 - 49	43,740	3,880

Calculate

a) General Fertility Rate

b) Age specific fertility rates

b) Total fertility rate

d) Crude Birth Rate

Chapter 7

7.0 Mortality

At the end of the module the student should be able to:

a) Define mortality

b) Calculate various measurements in mortality

c) Calculate the standardization of death rates

7.1 Introduction

The acceleration of population growth in the developing countries of the world has not been due to a change in the birth rates; but to the great reduction in death rate following improvements life in the developing countries, the rate of natural increase was most unlikely to have been more, and may indeed have been much less than that of the more developed countries.

There is still considerable scope for reduction in death rates in the developing countries. If this should occur unaccompanied by any fall in birth rates; the populations of these countries would grow at even faster rates than at present. In short, there is no hope of any reduction in the rates of population growth until the fall in birth rates is faster than that of the rates of death. In any case, the population will go on increasing until or unless the Crude Birth Rate (CBR) = Crude Death Rate (CDR) when the natural increase becomes zero.

7.2 Measurements of Mortality

To study mortality, it is necessary to use comparative measures or indices of mortality for periods and areas under consideration. The basic measures are essentially rates of incidence relating to the number of death to a unit of population, commonly 1,000, in a particular interval of time. The interval is invariably one year, to avoid the complicated effect of seasonal mortality variations on comparability of rates.

7.2.1 Crude Death Rate

This is the simplest measure, indicating the number of deaths in a particular year per 1,000 of the population.

Formula: - $CDR = \frac{D}{P} \times 1{,}000$

Where D = Deaths in a year

P = population at mid- year

The mid- year population is used, ideally, as the best available approximate of the average number of people exposed to death, or “at risk”, during the year. The term “crude” is used simply, since no allowance is made for often considerable differences between populations in those aspects of composition in particular age and rates increase with age from late childhood, and that at nearly all ages mortality is higher for males than females. Consequently, age and sex structure plays an important role in influencing crude death rates, regardless of factors like living standards and health programmes. This is why care has to be taken in interpreting international difference in crude death rates. There are some less developed countries (e.g. Costa Rica, Guyana, Jamaica, Malaysia, Sri Lanka) which have lower crude death rates (about 5-7 per 1,000), countries of North America and North Western Europe (about 9-12 per 1,000). Clearly, this is due to much younger age structures in less developed countries, resulting from their high fertility in the immediate past.

7.2.2 Age and sex – specific Death Rates

A more critical appreciation of mortality level can be derived from a table of mortality desegregated by age and sex. An age and sex specific death rate is the number of deaths during the year of persons of a given age and sex per 1,000 people of that age group and sex category.

Formula: A and $SSDR = \frac{D\ (A\&S)}{P\ (A\&S)} \times 1{,}000$

D (A&S) = Deaths of a given age and sex in a year

P (A&S) = population of a given age and sex in the middle of the year.

It is conventional to use five year or ten year age groups, as a compromise between unnecessarily detailed and tediously compiled larger age groups. However, because of the importance of infant mortality, separate rates are usually calculated for the under one and 1-4 age groups.

Examples of age and sex- specific death rates are given in the following table and figures for a developed and a less developed country. Although levels of mortality differ between England and Wales, and Mauritius, the overall pattern of mortality by age and sex is very similar. Mortality rates fall from high levels in the first one year of life to a minimum in 5-14 age group and then rise in successive age groups, although the mortality level in the first one year is not exceeded until old age. The lower mortality rate for females is found in all the England and Wales age groups and almost all in Mauritius. Traditional explanations include the greater exposure of men, in general to warfare, occupational and recreational hazards and to occupational stress. The excess male mortality at infant and even fetal stages suggests that biological difference is the decisive factor.

Table 5 Age Specific Mortality Rate in a District in Kenya

Age group	Population	No. of Deaths	ASDR
0 – 4	155,900	4108	26.35
5 – 14	498,100	6566	13.18
15 – 24	495,000	5252	10.61
25 – 34	487,400	1418	2.91
35 – 44	381,500	1823	4.78
45 – 54	322,300	3408	10.57
55 – 64	266,800	7022	26.32
65 – 74	143,300	9298	64.88
75 +	84,500	1325	15.68
	2,834,800	40220	

$$\text{Crude Death Rate} = \frac{\text{Deaths occurring during a calendar period}}{\text{Mid year population}} \times 1000$$

$$CDR = \frac{40220 \times 1000}{2834800} = 14.18795$$

7.2.3 Overall pattern of mortality rates

Sets of age – and sex specific rates represent refined measures of mortality, but they are cumbersome and difficult to assimilate. It is desirable, therefore to have a summary measure which epitomizes a whole set of rates but which avoids the crudity of the crude death rate. Demographers have used their technical skills to produce several measures and two of the most widely used will now be discussed to illustrate the procedure known as standardization. By this procedure, death rates are adjusted or standardized for the age and sex composition of the population. Standardization can be regarded as a means of holding age and sex constant.

7.3 Standardized Death Rates and Indices

A legitimate desire is often felt for a single mortality rate, summing up the rates to be made between one rate and another. For this purpose, standardized death rate is generally used. This is a hypothetical figure which indicates what the crude death rate would be if the population being studied had the same age and sex composition as a population (any real or assumed population).

You will find that while comparing rates between different geographical areas, different hospitals, population or experimental groups; there are other factors which should be considered in assessing the reliability of a crude death rate e.g. many diseases, age specific death rates are higher for infants and very old people and low for children and young adults as already shown. Therefore, if one population has a high proportion of very old and very young persons, the crude death rate will usually be higher than that of a population consisting of mainly children and young adults.

The following data shows the number of deaths in males by age of a population in country A and the resultant death rates (hypothesized) and her immigrants in a particular year.

Table 6 Population of country A and Immigrants to country A (Males)

AGE GROUP	MALES POPULATION	NO. OF DEATHS	MALES OF IMMIGRANTS	NO. OF DEATHS
0 - 4	1,900,000	1,406	26,000	21
5 -14	3,100,000	186	30,000	2
15 - 44	9,400,000	1,786	127,000	27
45 - 64	4,900,000	7,350	25,000	42
65 +	2,000,000	17,400	5,000	48
TOTAL	**21,300,000**	**28,128**	**213,000**	**140**

Table 7 Age Specific Death Rates for country A and Immigrants

AGE GROUP	DEATH RATE MALES OF COUNTRY A PER 100,000	DEATH RATE OF IMMEGRANTS PER 100,000
0 - 4	74.0	80.8
5 -14	6.0	6.7
15 - 44	19.0	21.3
45 - 64	150.0	168.0
65 +	870.0	960.0
C.D.R.	**132.1**	**65.7**

The crude death rate for country A is 132.1/100,000, and that of the immigrants is 65.7/100,000, although at every age group, age specific death rates were higher for immigrants than for country A. These results are explained by the relatively small number of people aged over 45 in country A, which gives much weight to the crude death rate. Here then is an example of crude death rate

being a sample of misleading index of comparing the levels of death in these two populations. Death rates vary with many social conditions, age and many others. To make a single figure index of reliable we must take account of the different population structures. This is what we have called a “standardized death rate”. There are two common methods of deriving standardized rates, known as direct and indirect standardization.

7.4 Direct Method of the Standardization

In order to get over the problem of different population structure of the two populations to be compared, we choose a standard comparison. This we call the standard. The population to be standardized we call it the index population. We then apply the age (or characteristics) specific rates of the index population to the standard population. This gives us the number of cases for each age group that we would expect if the index population rates applied in the standard population. Adding these cases over the age groups and dividing by the number of persons in the standard population gives the standardized death rate for the index population. In detail the steps involved in calculating the direct standardized death rate for the index population.

Define the standard population

Apply the Age Specific Death Rates for the index population to the numbers in each age group of the standard population to obtain the deaths we would expect in the population if the index rates applied.

Add these expected deaths over the age groups to obtain the total number of the expected deaths.

The crude rate obtained by dividing the total expected deaths by the total in the standard population; gives the standardized death rate for the index population.

Table 8 Standard Population and Expected Deaths of Country A against the Immigrants (per 100,000)

AGE GROUP	STANDARD POPULATION COUNTRY A	INDEX RATES IMMIGRANTS PER 100,000	EXPECTED DEATHS
0-4	1,900,000	80.8	1,536
5-14	3,100,000	6.7	208
15-44	9,400,000	21.3	2,002
45-64	4,900,000	168.0	8,232
65+	2,000,000	960.0	19,200
CDR	21,300,000	**146.4**	31,177

Table 9 Immigrants Population and Expected Deaths

AGE GROUP	IMMIGRANTS	INDEX RATES IMMIGRANTS PER 100,000	EXPECTED DEATHS
0 - 4	26000	74	19.24
5 -14	30000	6	1.8
15 - 44	127000	19	24.13
45 - 64	25000	150	37.5
65 +	5000	870	43.5
CDR	213000	**59.2**	126.17

The expected deaths are obtained by multiplying the standard population in each age group by the age specific death rate of the index population, e.g.

$$\frac{1{,}900{,}000 \times 80.8}{100{,}000} = 1{,}535$$

The total expected deaths are 1,535+208+2,002+8,232
+19,200 = 31,177

The standardized death rate for the immigrants is therefore:

$$\frac{31{,}177 \times 100{,}000}{21{,}300{,}000} = 146.4/100{,}000$$

The standardized death rate for country a is equal to the crude death rate which is 132.1 per 100,000.

In the following table are the standardized age- sexes – specific death rates for England and Wales, and Mauritius, which we came across earlier in table 4 using the population of Japan as a standard population to calculate the direct method of standardization.

The standardized death rates are as follows:

Mauritius

$$\text{SIR} = \frac{\text{Expected deaths}}{\text{Standard population}} \times 1{,}000$$

$$= \frac{518{,}314+445{,}676}{121{,}008{,}000} \times 1{,}000$$

$$= 7.97 \text{ per } 1{,}000$$

7.5 Choice of the Standard Population

Whatever considerations are made in the choice of a standard population for application of the direct method, the final decision as to which population is used must be to some extent arbitrary. However, provided the standard population is in some sense appropriate for the problem at hand, the precise structure of the

population is unlikely to affect the results of comparisons of an index population to a market extent. For example, if the problem is to compare several occupational groups in England and Wales an appropriate standard might be the adult population of Kenya. On the other hand, there is no good reason why the adult population of Kenya should be chosen as a standard if the problem is to compare several occupational groups in England and Wales an appropriate standard might be the adult population of Kenya. On the other hand, there is no good reason why the adult population of Kenya should be chosen as a standard if the problem is to make international comparisons, various suggestions have been made for the choice of standard populations and these are shown in the following table.

These standards vary from an African country with a low proportion of old people through an intermediate world population to a European population with a high proportion of old people.

Table 10 Standard Population used for International comparison

Age Group	Africa	World	Europe	Truncated
0	2000	2400	1600	
1 – 4	2000	9600	6400	
5 – 9	10000	10000	7000	
10 – 14	10000	9000	7000	
15 – 19	10000	9000	7000	
20 – 24	10000	8000	7000	
25 – 29	10000	8000	7000	
30 – 34	10000	6000	7000	
35 – 39	10000	6000	2000	6000
40 – 44	5000	6000	2000	6000
45 – 49	5000	6000	2000	6000
50 – 54	3000	5000	7000	5000
55 – 59	2000	4000	6000	4000
60 – 64	2000	4000	5000	4000

65 – 69	1000	3000	4000	
70 – 74	1000	2000	3000	
75 – 79	500	1000	2000	
80 – 84	300	5000	1000	
85 +	200	5000	1000	

7.5.1 Indirect standardization

The direct method of standardization uses the population of each age group as the standard. The indirect method of standardization uses the age specific rates as the standard. In detail, the procedure is as follows:

Define the standard age specific mortality rates.

Apply the standard age specific rates to the index population to get the expected deaths in each age group.

Add the expected deaths of all the age groups to get the sum of the expected deaths.

Divide the total observed deaths (real) by the total of the expected deaths to get the standardized mortality ratio

Multiply the standardized mortality ratio by the crude death rate of the standard population.

Applying the procedure mentioned above to the data shown in the following table the calculation would be as follows:

Table 11 Indirect Standardization of the Immigrants to Country A

Age Group	Standard Rate/1000	Immigrant population	Expected Cases
0 – 4	74.0	26000	19.2
5 – 14	6.0	30000	1.8

15 – 44	19.0	127000	24.1
45 – 64	150.0	25000	37.5
65 +	870.0	5000	43.5
			126.1

The expected cases are obtained by multiplying the age specific rates by the number of persons in the immigrants (index) population e.g. (74.0 x 26,000) divide by 100,000 = 19.2.

The standard rates are the age specific rates for country A. The total expected cases are arrived at by adding 19.2+ 1.8+ 24.1+ 37.5 + 43.5= 126.1, and hence the standard mortality ratio

(SMR) is $\frac{140}{126.1}$ =1.11 or 111%

Where 140 are the actual or observed total number of deaths in the immigrants population.

The indirectly standardized rate for the immigrants is then obtained by multiplying the crude rate of the standard population by the standard mortality ratio that is 132.1 x 1.11 = 146.6 per 1,000. The indirect standardized rate for country A remains 132.1 for purposes of comparison, because we chose country A's rates as the standard.

7.5.2 Choice between the Direct and Indirect Methods

Advantages of the direct method:

The method is simple to understand and apply

If it is necessary to compare one index population with another index population, it is possible to simply take the ratio of the two comparative mortality indices, to obtained comparative mortality index

$$CMI = \frac{\text{DIRCTLY STANDARDIXED RATE 1}}{\text{DIRECTLY STANDARDIZED RATE 2}}$$

The ratio of the two SMR obtained through indirect standardization does not simplify to any meaningful expression. Indeed in extreme situations the ratio of the two Standardized Mortality Rates (SMRs) could yield a totally misleading comparison of the incidence/mortality in the two index population.

Advantages of the indirect method:

It can be shown that assuming the index rates are small as indeed they are in most studies of mortality and incidence the standard error of indexing the indirectly standardized rates is less than the directly standardized rate. Thus, if the sample sizes in the index population are small, there may be an important gain in precision if the indirect method is used rather than the direct method.

In many situations, it may be impossible to apply the direct method because the distribution of cases or deaths in the index population is unknown. The indirect method does not require this information.

In summary, the indirect method is preferred because of its consistency provided the sample sizes in the index population are small.

Questions for the module

1. The following information was collected from Meru District in 1997

Age group	Population	No. of Deaths
0 – 4	157,900	3108
5 – 14	598,100	5566
15 – 24	475,000	4252

25 – 34	387,400	1383
35 – 44	281,500	1623
45 – 54	212,300	2408
55 – 64	176,800	6022
65 – 74	133,000	8298
75 +	74,500	1225

Using the information provided in the table, calculate: -

i) Crude death rate ii) Age specific death rate

iii) Graph the age specific death rate iv) Comment on the graph.

2) Briefly describe the following rates commonly used in demography

i) Infant Mortality Rate ii) Neonatal Mortality Rate

iii) Post neonatal Mortality Rate iv) Fetal Death Rate

v) Maternal Mortality Rate vii) Case Fatality Rate

Chapter 9

8.0 Introduction

At the end of the module the student should be able to:

a) Define the age structure of the population
b) Describe age misreporting
c) Describe age heaping
d) Describe age shifting
e) Calculate and interpret whipples index

8.1 The determinants of age structure

The age structure and age distribution in any given population is entirely determined by the levels of mortality, fertility and migration occurring in the past. These three (3) factors affect distribution in different ways and to different extent. Changes in the levels of fertility will at first affect just younger age groups only over many years will it work through to affect older ages. The effect of fertility changes can thus be very dramatic, especially if the rise or fall is rapid.

Variations in the level of infant mortality tends to produce an effect very similar to that caused by change in fertility affecting initially the young ages, and only gradually working through to the whole age range.

A change in adult mortality has however, generally, a much less dramatic impact on the shape of an age distribution that either fertility or infant mortality. This is because it is largely not concentrated in one age group but is spread over all ages.

Migration can cause big distortion in age distributions because it is normally concentrated among young adults and is sometimes also sex selective. Areas from which substantial numbers have emigrated will in extreme cases have top heavy pyramids with a high proportion of the proportion in older age groups, few young adults and consequently, few children conversely areas where substantial numbers of immigrants have settled will have a young age structure, with many children and very few elderly.

Overall therefore, an age distribution is best regarded as being determined primarily by fertility and modified, to a greater or lesser extent, by mortality and migration.

8.2 Age Misreporting

Distortion in age distribution may be real and is caused by past changes in the levels or patterns of mortality, fertility or migration or they may be due to errors. The errors may be caused either by some individuals being incorrectly omitted from a survey or by ones individuals ages being wrongly reported.

A great problem for formal demography, frequently prohibiting the use of conventional techniques is that people simply do not tell the truth.

This is not a problem in developed countries but in many parts of the world, age misreporting can take two basic forms:-

- Age heaping
- Age shifting

Age Heaping

Age heaping or digit preference as it is often called. Too many people report ages ending with 0 or 5 while relatively few people give ages ending in 9, 1, 4 or 6. For example, if in this group ones age is 22, the individual will report this age as 20 years or 25 years.

Age Shifting

Age shifting is a more serious problem than age heaping largely because it is harder to detect and adjust. Typically older people tend to exaggerate their ages, particularly where age confers status. Young men may underrate their age to avoid state free services. While older unmarried women may underrate their age. Numerous techniques exist both to estimate the extent to which ages are misreported and to correct age distributions. Techniques for adjusting and smoothing ages are often complex and will not be discussed at this level. Whipple's index is a measure of preferences for ages ending in 0 and 5. It takes ages between 23 and 62 because outside this range shifting and other problems always tend to confuse the normal pattern of heaping. The index is calculated simply by expressing the population reporting age ending with 0 and 5 as a percentage of the total population age between 23 & 62 and multiplying by 5. The range of the index is then from 100 indicating no preference for 0 and 5 up to 500 indicating that only 0 and 5 were reported.

Table 12 Whipples Index table of interpretation

Key	
Quality	**Whipples index**
Higher accurate	Under 105
Fairly accurate	105 – 110
Approximate	110 - 125
Rough	125 - 175
Very rough	Above 175

Exercise

(a) Calculate the Whipples index for Wajir population

The whipplies index measures preferences for ages ending in 0 or 5 as a percentage of the total population aged between 23 and 62 and multiplying by 100 and then by 5.

FORMULAR

$$= \frac{\text{Population aged } 25+30+35+40\ldots\ldots+60}{\text{Population aged between 23 – 62 years (inclusive)}} *5*100$$

Table 13 Whipples index data for Wajir population

AGE	MALES	FEMALES	TOTALS
25	257	263	520
30	336	407	743
35	140	188	328
40	20	319	525
45	109	97	206
50	175	256	431
55	45	68	113
60	96	171	267
TOTALS			3133

Table 14 Ages between 23 and 62

AGE	MALES	FEMALES	TOTALS
23	133	106	239
24	163	153	316
25-29	703	773	1476
30-34	666	652	1318
35-39	383	409	792

40-44	361	539	900
45-49	246	184	430
50-54	271	417	688
55-59	124	125	249
60-61	96	171	267
62	28	13	41
Total			6716

TOTAL = 6716

=3133/6716*5*100

=233.24895771

=233.25 WHIPPLES INDEX

Interpretation = Too many people reported their ages ending with 0 or 5 “VERY ROUGH”

b) Construction of age pyramid

Table 15 Age Pyramid

AGE	MALES	FEMALES
0-4	1601	1529
5-9	1499	1490
10-14	1457	1275
15-19	1049	1055
20-24	793	798
25-29	903	773
30-34	666	652
35-39	383	409
40-44	361	439

45-49	246	184
50-54	271	317
55-59	124	125
60-64	143	208
65-69	58	64
70-74	89	113
75-79	34	47
80-84	66	157

Table 16 Age and Sex Structure of Bangladesh in 1970

AGE	MALE	FEMALE
0-4	7457	7192
5-9	5820	5625
10-14	4730	4533
15-19	4017	3777
20-24	3362	3101
25-29	2891	2661
30-34	2391	1793
35-39	1983	1484
40-44	1635	1222
45-49	1343	974
50-54	1066	784
55-59	858	601
60-64	658	440
65-69	467	304
70-74	339	334
75+	339	
TOTAL	393556	34825

THE WHIPLES index for Bangladesh

Question of the module:

Age misreporting can happen due to various factors. Discuss those factors and show how the Whipples index is used to detect the digit preference for ages ending in 0 and 5.

Chapter 9

9.0 Introduction

At the end of the module the student should be able to:

a) Define a life table
b) Describe the uses of a life table
c) Calculate the life table

9.1 LIFE TABLE

A life table is a life history of a hypothetical group or cohort of people as it is diminished gradually by deaths.

<u>The Uses Of A Life Table</u>

The construction and handling of a life table are matters of professional demographers.

Health and rural planners should however at least to be aware of the data that can be obtained from life tables so that they may be applied usefully in the course of their work.

In particular they should be aware that a life table can show:-

- ✓ The probability of dying within some specific time period of persons within some age groups.
- ✓ The average number of years a new born infant can be expected to live
- ✓ The average number of years remaining for any person of any age.
- ✓ The probability of surviving for any given age to any other age.
- ✓ The probability of surviving for any given number of years for persons of any age.

Terms Used In Life Table

dx = the deaths during a year of persons aged x

px = mid year population

nmx = ASMR age specific mortality rate

nqx = probability of dying between exact ages x and x+1

npx = probability of surviving between exact age x and x+1

nax = proportion of the year lived by those who died.

lx = life table population.

ndx = life table deaths

nlx = exact person years lived, proportion of the year lived by those who died.

tx = total person years lived after x

ex = life expectancy

THE FORMULAS

$$nmx = \frac{dx \text{ (death of persons aged x)}}{px \text{ (population of persons aged x)}}$$

nax = proportion of the year lived by those who died and it is always given.

$$nqx = \frac{n.\ nmx}{1 + n\ (n-nax).nmx}$$

Where n= class interval

Calculations of nqx

$$nq1 = \frac{n.nmx}{1+n\ (1-nax).nmx} = 0.0207113$$

$$= \frac{1*0.021016}{1+1(1-\ 0.3)*0.021016}$$

$$nq4 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{4*0.016549}{1+4(1-0.4)*0.016549} = 0.063667$$

$$nq9 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.011396}{1+5(1-0.5)*0.011396} = 0.055402$$

$$nq14 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.009341}{1+5(1-0.5)*0.009341} = 0.045639$$

$$nq19 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.006009}{1+5(1-0.5)*0.006009} = 0.0296003$$

$$nq24 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.01}{1+5(1-.05)*0.01} = 0.04878$$

$$nq29 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.001983}{1+5\,(1-0.5)*0.00198} = 0.0098$$

$$nq34 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.004974}{1+5(1-0.5)*0.004974} = 0.0245499$$

$$nq39 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.008656}{1+5(1-0.5)*0.008656} = 0.4236325$$

$$nq44 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.001221}{1+5(1-0.5)*0.001221} = 0.0060864$$

$$nq49 = \frac{n.nmx}{1+n\,(1-nax).nmx} = \frac{5*0.009355}{1+5(1-0.5)*0.009355} = 0.045706$$

$$nq54 = \frac{n.nmx}{1+n\,(1-nax).nmx}$$

$$=\frac{5*0.002816}{1+5(1-0.5)*0.002816}$$

$$=0.013982$$

$$nq59 = \frac{n.nmx}{1+n\ (1-nax).nmx}$$

$$=\frac{5*0.002846}{1+5(1-0.5)*0.002846}$$

$$=0.0141294$$

$$nq64 = \frac{n.nmx}{1+n\ (1-nax).nmx}$$

$$= \frac{5*0.024363}{1+5(1-0.5)*0.024363}$$

$$=0.1148215$$

$$nq69= \frac{n.nmx}{1+n\ (1-nax).nmx}$$

$$= \frac{0.003937*5}{1+5(1-0.5)*0.003937}$$

$$=0.0194931$$

$$nq74= \frac{n.nmx}{1+n\ (1-nax).nmx}$$

$$= \frac{0.477941*5}{1+5(1-0.5)*0.477941}$$

$$=1.088777$$

npx = 1 - nqx (the probability of surviving)

lx = Start with a rounded figure – assuming = 100,000

= 100,000* chances of children surviving

= 100000*npx

= npx * lx (upper)

eq. TABLE lx = npx * lx (upper)

First lx is given i.e for (0 -1) age group = 100,000

1 - 4 (age) = npx (1st class * lx 1st class)

= 0.979289*100000

= 97929

Therefore l_4= 97929

ndx = nqx * lx (upper)

= 100,000 *nqx or (nqx *lx)

(0-4) = 020207113*100000

= 2071

nlx = $\frac{(lo+l1)i}{2}$

= $\frac{lx(lower)+lx(upper) * 5 \text{ (Class interval)}}{2}$

= $\frac{lx\ (upper) + lx\ (lower)}{2} *5$

= $\frac{(100000+97929)*5}{2}$

= 494823

nlx last class i.e 85+ = $\frac{ndx\ (last)}{nmx\ (last)}$

To get the last figure for the last class in the nlx column, you get figures for the last ndx and nmx for the last class and divide as above. The figure you will get is added cumulatively from the bottom i.e. the last figure of column nlx thus forming column Tx as shown on chart.

Tx = Cumulative total from bottom of nlx

= cumulatively down –up

= cumulative from last class nlx upwards eg. 118161+282370-400531

Expectancy is then calculated by dividing Tx/lx of each age group

Ex = $\frac{Tx}{Lx}$

= 6116560/100000

= 61.1

= 61

(e.g.)

0<1 = 6649664/100000 = 66.4 = 66

1-4	=	Tx (row 1-4)/lx (row 1-4)	=	6166596/93227	=	66
5-9	=	Tx (row 5-9)/lx (row 5-9)	=	5709774/92080	=	62
85+	=	Tx (last row 1-4)/lx (last row 1-4)				
	=	30625/7990 = 3.8 = 4				

AGE GROUP	Px	dx	nmx	nax	Nqx	npx	lx	ndx	nlx	Tx	ex
0 -1	3759	79	0.021016	0.3	0.020711	0.97929	100000	2071	494822	6096588	61
1 – 4	7130	118	0.016550	0.4	0.063633	0.93637	97929	6232	474066	5601766	57
5 – 9	10179	116	0.011396	0.5	0.055402	0.9446	91697	5080	445786	5127700	56
10 -14	8885	83	0.009342	0.5	0.045639	0.95436	86617	3953	423203	4681914	52
15 -19	7322	44	0.006009	0.5	0.0296	0.9704	82664	2447	407203	4258711	51
20 - 24	6600	66	0.010000	0.5	0.04878	0.95122	80217	3913	391303	3851508	48
25 - 29	6051	12	0.001983	0.5	0.009866	0.99013	76304	753	379639	3460205	45
30 - 34	4425	22	0.004972	0.5	0.02455	0.97545	75551	1855	373120	3080566	41
35 - 39	3350	29	0.008657	0.5	0.042363	0.95764	73697	3122	360678	2707446	38
40 - 44	2456	3	0.001221	0.5	0.006086	0.99391	70575	430	351799	2346768	33
45 - 49	1920	18	0.009375	0.5	0.045802	0.9542	70145	3213	342693	1994969	28
50 - 54	1420	4	0.002817	0.5	0.013986	0.98601	66932	936	332321	1652276	25
55 - 59	1054	3	0.002846	0.5	0.014129	0.98587	65996	932	327650	1319955	20
60 - 64	903	22	0.024363	0.5	0.114822	0.88518	65064	7471	306642	992305	15
65 - 69	508	2	0.003937	0.5	0.019493	0.98051	57593	1123	285158	685663	12
70 - 74	345	0	0.000000	0.5	0	1	56470	0	282352	400505	7
75+	272	130	0.477941	0.5	1	0	56470	56470	118153	118153	2

Table 17 Presentation of life table Table 18 Life table presentation

AGE GROUP	px	dx	nmx	nax	nqx	npx	lx	ndx	nlx	Tx	ex
0<1			0.0711	0.3	0.067729	0.932271	100000	6773	483068	6656110	67
1-4			0.0031	0.4	0.012308	0.987692	93227	1147	463268	6173042	66
5-9			0.0014	0.5	0.006976	0.993024	92080	642	458795	5709774	62
10-14			0.0011	0.5	0.005485	0.994515	91438	502	455935	5250979	57
15-19			0.0016	0.5	0.007968	0.992032	90936	725	452868	4795044	53
20-24			0.0024	0.5	0.011928	0.988072	90211	1076	448365	4342176	48
25-29			0.0025	0.5	0.012422	0.987578	89135	1107	442908	3893811	44
30-34			0.0032	0.5	0.015873	0.984127	88028	1397	436648	3450903	39
35-39			0.0040	0.5	0.019802	0.980198	86631	1715	428868	3014255	35
40-44			0.0057	0.5	0.028100	0.971900	84916	2386	418615	2585387	30
45-49			0.0096	0.5	0.046875	0.953125	82530	3869	402978	2166772	26
50-54			0.0113	0.5	0.054948	0.945052	78661	4322	382500	1763794	22
55-59			0.0186	0.5.	0.088868	0.911132	74339	6606	355180	1381294	19
60-64			0.0275	0.5	0.128655	0.871345	67733	8714	316880	1026114	15
65-69			0.0425	0.5	0.192090	0.807910	59019	11337	266753	709234	12
70-74			0.0652	0.5	0.280310	0.719690	47682	13366	204995	442481	9
75-79			0.1034	0.5	0.410807	0.589194	34316	14097	136338	237486	7
80-84			0.1734	0.5	0.604813	0.395137	20219	12229	70523	101148	5
85+			0.2609	0.5	1	0	7990	7990	30625	30625	4

SOURCE DOCUMENTS OF DEMORAPHIC INFORMATION

Census reports – Census Bureau of Statistics (CBS)

Census projections – Reports 1980-2000 (CBS)

Demographic Survey – 1977 CBS

Kenya fertility survey – 1977, 1978, 1993

Statistical abstract

Statistical Digest – quarterly

UN statistical year book

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5. Gachanja, F.W (1989). *Health and Demography: A Case study of two antenatal clinics in Nairobi*- A Thesis

Mount Kenya University

Department of Clinical Medicine
Bsc in Health Records and Information Management
End of Term Examination.

Subject: Medical Demography Date: Time allowed: $2^1/_2$ hours

Instructions

1. Write your college number at the right hand of each answer sheet.
2. Begin each answer on a separate answer sheet
3. Answer all questions

Q1. From the following table relating to the number of women and births which occurred in Mukono District in Uganda in 1987. Population of the District was reported at 1,267,897

Calculate

a) General Fertility Rate

b) Age specific fertility rates

c) Crude Birth Rate

d) Total fertility rate

e) Interpret your answer in (d) above

Age groups	No. of women	No. of births
15 - 19	97,892	15,357
20 – 24	135,651	26,687
25 – 29	125,585	24,979
30 – 34	95,918	13,878
35 – 39	88,874	10,698
40 – 44	60,286	2,764
45 - 49	52,773	1,885

(30 Marks)

Q2. The following information was collected from Webuye District in 1996

Age group	Population	No. of Deaths
0 – 4	255,956	5177
5 – 14	598,156	7589
15 – 24	595,564	6288
25 – 34	587,457	2489
35 – 44	481,578	2827
45 – 54	422,387	4499
55 – 64	366,899	8066
65 – 74	243,879	10297
75 +	94,678	2356

Using the information provided in the table, calculate: -

i) Crude death rate ii) Age specific death rate

iii) Graph the age specific death rate iv) Comment on the graph.

(30 Marks)

Q3 Briefly describe the following rates commonly used in demography

i) Infant Mortality Rate ii) Neonatal Mortality Rate

iii) Post neonatal Mortality Rate iv) Maternal Mortality Rate

v) Case Fatality Rate

(10 Marks)

Mount Kenya University

Department of Clinical Medicine
Bsc in Health Records and Information Management
End of Term Examination.

Subject: Medical Demography Date: Time allowed: $2^1/_2$ hours

Instructions

1. Write your college number at the right hand of each answer sheet.
2. Begin each answer on a separate answer sheet
3. Answer all questions

Q1. The following data refer to cases of Malaria for Uganda population and Immigrants.

Age Group	Uganda Population (000)	Cases	Immigrants population (000)	Cases
0 – 4	2000	1505	26	31
5 – 14	4100	200	23	6
15 – 44	9500	1686	127	36
45 – 64	5800	7650	28	50
65 +	2000	18000	5	55

(a) Calculate the crude incidence rates for both Uganda's population and immigrants.

(b) Using the direct method of standardization, standardize the immigrant's crude rate using the population of Uganda as your standard.

(30 Marks)

Q2. a) Briefly describe the following ratios:-

i) Child Women Ratio

ii) Dependency Ratio

b) Show with example how sex ratio is calculated

c) Describe the implication of sex ratio derived from the calculations

d) Describe the shape of sex/age pyramid of the current population of Kenya.

(20 Marks)

Q3. Describe the consequences a country would face if it fails to educate its citizens of family planning.

(20 Marks)